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Re: Application for the EngD position – Developing an Interactive Reverse Logistics Map for Circular Construction (CIRCOLOGIC), University of Twente

Dear Professor Hans Voordijk, Professor Berkhout, Professor Marc, and Members of the Selection Committee,

The construction industry is gradually shifting from the traditional "take, make, use, and dispose" model towards a circular economy, where building materials are reused, refurbished, and recycled to extend their value and reduce waste. While digital technologies such as BIM, GIS, digital twins, and material passports have significantly improved how construction information is managed, one critical challenge remains: valuable materials often fail to re-enter the construction cycle because information about their availability, quality, location, and future demand is fragmented across multiple stakeholders. Reverse logistics has emerged as a key process for recovering and redistributing these materials, yet without integrated digital decision-support systems, circular construction cannot reach its full potential. This is why the idea of developing an interactive reverse logistics map immediately captured my interest.

My interest in this area developed gradually. During my Master's in Environmental Engineering, my thesis, Agile Project Management in the Construction Industry, explored how Agile methodologies improve collaboration, continuous risk management, adaptive decision-making, and resource efficiency in construction projects. Through this research, I realised that successful transformation in construction depends not only on technical innovation but also on effective coordination, transparent information flow, and collaboration between stakeholders.

Completing my thesis led me to ask a broader question: What happens to construction materials once a building reaches the end of its service life? As I explored circular construction, I realised that one of the biggest barriers to material reuse is not simply recovering materials but ensuring that reliable information about those materials is available to everyone who needs it. Without visibility into material availability, quality, ownership, location, and logistics, reusable resources often become waste. That realization shifted my interest from improving project delivery to designing digital solutions that enable circular decision-making across the entire construction value chain.

The Engineering Doctorate position Developing an Interactive Reverse Logistics Map for Circular Construction (CircoLogic) at the University of Twente immediately stood out to me because it addresses exactly this challenge. Rather than focusing solely on research, the project aims to develop a practical decision-support platform that integrates geospatial information, reverse logistics, and stakeholder collaboration to make circular construction more achievable in practice. I find that combination of sustainability, digital innovation, and industrial implementation both technically challenging and highly meaningful.

I believe my professional experience has prepared me well for this interdisciplinary project. Working on digital twins gave me experience in organising and integrating information from physical assets into digital decision-support systems. At Hive CPQ, I managed large datasets, gathered requirements from manufacturers and customers, and developed configurators that transformed complex rules into intuitive digital tools. Those experiences taught me that the success of any digital platform depends not only on technology but also on understanding users, structuring reliable data, and translating stakeholder needs into practical solutions.

In addition, consulting construction companies on Agile implementation strengthened my understanding of collaboration across complex stakeholder networks. Agile promotes continuous communication, iterative improvement, and adaptability, principles that are equally valuable in circular construction, where successful material reuse depends on effective coordination between demolition contractors, designers, logistics providers, recyclers, municipalities, and clients. Combined with my background in civil engineering, environmental engineering, and GIS, I believe I can contribute not only to the technical development of the interactive logistics platform but also to designing a solution that stakeholders will genuinely adopt and use.

What attracts me to the University of Twente is its strong focus on solving real societal challenges through engineering and its close collaboration with industry. The structure of the EngD programme, with its balance between advanced education and an industry-driven design project, is exactly the environment in which I want to continue developing. I enjoy learning new technologies and methods, but I find the greatest value in applying them immediately to solve real-world problems. The opportunity to work with researchers, industry partners, and organisations such as Dura Vermeer and TNO while developing an innovative digital solution makes this programme particularly appealing.

Looking ahead, I want to contribute to the digital transformation of the construction industry by designing technologies that make circular construction more practical, data-driven, and widely adopted. For me, the transition towards a circular economy is not only a sustainability challenge but also an information and decision-making challenge. I believe digital technologies can bridge that gap, and I see this EngD as the ideal opportunity to develop into an engineer who connects technology, sustainability, and industry to create solutions that deliver measurable impact.

I would be honoured to contribute to the CircoLogic project and would welcome the opportunity to discuss how my background, professional experience, and motivation align with the objectives of this Engineering Doctorate programme.

Thank you for considering my application.

Yours sincerely,

Pooja Awasare